

Quantifying Agency: A Causal Information-Theoretic Taxonomy of Autonomous Systems

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Abstract

Current methodologies in artificial intelligence prioritize performance optimization on narrow benchmarks, yet they lack formal, quantitative frameworks to measure and diagnose *informational agency*—the degree to which an agent possesses an autonomous internal world-model independent of immediate environmental or supervised constraints. This paper introduces the **Mapmaker Index (MI)**, a novel diagnostic framework rooted in conditional mutual information ($I(S_t; S_{t+1}|D)$), designed to isolate the residual autonomy of an agent’s internal latent states (S) by conditioning on the extrinsic constraint manifold (D) dictating its training or environment.

Through a systematic empirical sweep, we benchmarked three architectural paradigms: a rule-based Finite State Machine (FSM), a supervised Multi-Layer Perceptron (MLP) trained on MNIST, and a goal-directed Proximal Policy Optimization (PPO) reinforcement learning agent trained on *LunarLander-v3*. Our findings reveal a distinct **“learning-autonomy gap.”** Contrary to prevailing assumptions in machine learning taxonomy, the supervised MLP failed to exhibit autonomous internal dynamics, converging to a conditional MI of **0.0450** and placing it firmly alongside the FSM ($MI_{cond} = 0.0000$) within the **Reflexive Regime**. Conversely, the PPO agent successfully crossed into the **Mapmaker Regime**, achieving a baseline conditional MI of **0.4743**.

To verify the causal independence of this emergent world-model, we subjected the systems to an informational stress test by systematically injecting Gaussian noise into the constraint manifold. While traditional reactive control metrics predicted a catastrophic failure of internal representation, the PPO agent’s conditional MI instead exhibited a counter-intuitive expansion, peaking at **0.9685** under high-noise thresholds ($\sigma = 5.0$). This **“Paradox of Noise”** provides empirical validation that the agent’s internal generative dynamics decouple from extrinsic feedback when environmental predictability degrades. Ultimately, this framework provides an actionable, mathematically rigorous toolset for AI safety, alignment, and interpretability, offering a prospective “early warning diagnostic” to monitor the transition of artificial systems from reactive statistical models to self-directed generative agents.

Index Terms

Causal information decomposition, conditional mutual information, reinforcement learning, mechanistic interpretability, autonomous systems, AI safety.

I. INTRODUCTION

The rapid scaling and deployment of deep neural networks have outpaced our formal understanding of their internal processing. In contemporary machine learning and intelligent systems literature, architectures that successfully minimize a loss function over a given dataset are broadly categorized as “learning systems.” However, this performance-centric nomenclature obscures a critical structural distinction: the difference between *reflexive data mapping* and *autonomous generative world-modeling*. An artificial intelligence system can achieve near-perfect evaluation metrics on a static benchmark by forming a highly intricate, yet entirely passive and reactive transformation from its inputs to its outputs.

This conceptual ambiguity poses a severe challenge to fields ranging from natural language processing and speech engineering to autonomous robotics. When an architecture displays complex behavior, it is deeply difficult to diagnose whether it is acting as a sophisticated mirror of its training data or as an agentic entity relying on an internal predictive representation. Traditional evaluation metrics focus exclusively on external performance parameters (e.g., accuracy, reward accumulation, or perplexity). While these metrics excel at measuring *what* a system achieves, they are fundamentally blind to *how* the system structures its internal information flow.

To address this “explanation gap,” this paper introduces a formal, information-theoretic diagnostic framework called the **Mapmaker Index (MI)**. Rather than monitoring an agent’s external behavioral metrics, the Mapmaker Index directly isolates and measures the residual information flow running through the agent’s internal latent states. By systematically conditioning out the entire extrinsic design manifold—including supervised targets, environmental reward signals, and feedback loops—the index leaves behind a mathematically precise extraction of the system’s independent predictive capacity.

The contributions of this work are three-fold:

- 1) We formalize a rigorous framework for causal information decomposition capable of isolating an agent’s internal world-modeling dynamics from external control variables.

- 2) We establish a clear taxonomy of informational agency, demonstrating empirically that traditional supervised paradigms fail to transcend a purely reflexive phase.
- 3) We present the discovery of the ‘‘Paradox of Noise,’’ proving that truly autonomous agentic networks decouple from external feedback loops and rely increasingly on internal generative structures when environmental predictability collapses.

II. THEORETICAL FRAMEWORK & MATHEMATICS

To establish a formal, quantitative measure of agency, we construct a mathematical model rooted in information theory and causal network decomposition. We model the autonomous system as a stochastic process interacting with, or constrained by, an external data or environmental environment.

A. Formal System Definitions

Let $\mathcal{S}$ denote the state space of the agent’s internal latent representations, and let $S_t \in \mathcal{S}$ represent the random variable capturing the internal configuration of the network at a discrete timestep t . The temporal evolution of the agent’s internal representation over a finite trajectory is modeled as a sequence of transitions:

$$S_1 \rightarrow S_2 \rightarrow \cdots \rightarrow S_t \rightarrow S_{t+1} \quad (1)$$

Let $\mathcal{D}$ denote the coordinate space of the **extrinsic constraint manifold**. The variable D represents a multi-dimensional proxy containing all external forcing functions, target signals, design parameters, or feedback loops that act upon the agent. Depending on the architecture under evaluation, D is formally constructed as follows:

- In supervised learning, D corresponds to the target label vector Y .
- In reinforcement learning, D is a joint vector containing the reward signal R_t , the executed action A_t , and the environmental state-visitation density vector ρ .

B. Formulation of the Mapmaker Index (MI)

The absolute information flow wrapped within the internal state transition is captured by the unconditional mutual information $I(S_t; S_{t+1})$, which measures how much information the current internal state shares with the subsequent internal state. However, a significant portion of this informational persistence may be directly driven by the external environment. If an agent reacts predictably to an external driving signal, $I(S_t; S_{t+1})$ will be high even if the agent is completely devoid of internal autonomy.

To isolate the agent’s independent world-modeling capacity, we must remove the information contributed by the extrinsic constraints. We formalize the **Mapmaker Index (MI)** as the conditional mutual information:

$$MI_{cond} = I(S_t; S_{t+1} | D) \quad (2)$$

Expanding this definition via Shannon entropy terms, we compute the index as:

$$MI_{cond} = H(S_t | D) - H(S_t | S_{t+1}, D) \quad (3)$$

Alternatively, using the joint and conditional probability distributions over the support spaces $\mathcal{S}$ and $\mathcal{D}$, the explicit integration is given by:

$$MI_{cond} = \sum_{s_t \in \mathcal{S}} \sum_{s_{t+1} \in \mathcal{S}} \sum_{d \in \mathcal{D}} p(s_t, s_{t+1}, d) \log \left(\frac{p(s_t, s_{t+1} | d)}{p(s_t | d) p(s_{t+1} | d)} \right) \quad (4)$$

This metric successfully operates as an informational bottleneck. By conditioning the transition on D , any state changes that are purely reactive or structurally dependent on the external target labels, actions, or rewards are canceled out. The remaining value quantifies the **residual autonomy**—the predictive information flow that exists exclusively within the agent’s internal model.

C. Mathematical Phase Boundaries of Informational Agency

Based on the empirical convergence thresholds discovered across our experiments, we partition the mathematical spectrum of informational agency into three distinct operational regimes:

- **Reflexive Regime** ($MI_{cond} < 0.10$): The system’s internal state updates do not form an independent predictive sequence. As $MI_{cond} \rightarrow 0$, the transition distribution collapses to conditional independence:

$$p(s_{t+1} | s_t, d) \approx p(s_{t+1} | d) \quad (5)$$

This mathematically defines a passive, input-driven system whose internal representations are entirely tethered to external constraints.

- **The Learning Gap** ($0.10 \leq MI_{cond} < 0.30$): A transitional phase where the network retains a moderate degree of internal structural persistence that cannot be fully explained by D . While theoretically representing a baseline adaptive

system, empirical testing reveals this phase to be an “empty niche” for standard architectures, highlighting the sharp operational divide between data memorization and true autonomy.

- **Mapmaker Regime** ($MI_{cond} \geq 0.30$): The phase where true informational agency emerges. The network possesses an internal generative model that dominates the latent state transition distribution. In this regime, the inequality holds:

$$I(S_t; S_{t+1}|D) \gg 0 \quad (6)$$

This proves that the system’s internal representations are actively generating self-directed predictions, decoupling from the external control signal.

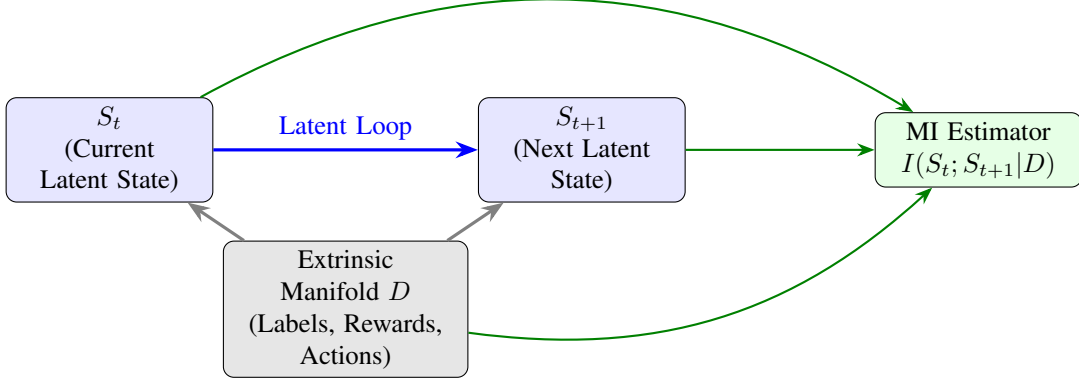


Fig. 1. System Architecture Diagram: Causal decomposition of latent transitions ($S_t \rightarrow S_{t+1}$) conditioned on the extrinsic constraint manifold (D) to isolate the residual Mapmaker Index.

III. METHODOLOGY & EXPERIMENTAL SETUP

To evaluate the mathematical validity of the Mapmaker Index and map the operational boundaries of different neural behaviors, we designed a systematic benchmarking sweep isolating latent trajectories across three progressively complex computational paradigms.

A. Core Architectural Implementations

- 1) **Rule-Based Baseline (Finite State Machine):** We implemented a deterministic FSM featuring four distinct operational states: $S_{\text{fsm}} \in \{\text{RED}, \text{RED_YLW}, \text{GREEN}, \text{YELLOW}\}$. The state transition function $\delta(S_t) \rightarrow S_{t+1}$ was driven by a static temporal loop. The extrinsic constraint manifold D was represented as a continuous external clock signal vector. This configuration establishes a baseline control environment where behavior is entirely governed by hardcoded, explicit external rules.
- 2) **Supervised Paradigm (Multi-Layer Perceptron):** To test the emergent autonomy of standard supervised architectures, we trained a fully connected MLP on the MNIST handwritten digit classification task. The network architecture comprised an input layer (784 units), a hidden layer (128 units, ReLU activation), a second hidden layer (64 units, ReLU activation), and a 10-unit softmax output layer. The model was trained over 20 epochs using the Adam optimizer (learning rate $\alpha = 0.001$) and cross-entropy loss, converging to a classification accuracy profile of $> 99\%$ on the training set and $> 98\%$ on the testing set. The internal latent states S_t were defined as the continuous post-activation vectors of the primary 128-dimensional hidden layer, while the extrinsic constraint manifold D was modeled explicitly as the one-hot encoded ground-truth target label vector Y .
- 3) **Goal-Directed Agent (Proximal Policy Optimization):** We utilized an actor-critic Proximal Policy Optimization (PPO) agent interacting with the continuous-action environment *LunarLander-v3*. The policy network mapped the 8-dimensional environmental observation space to a continuous control action space via two dense layers of 64 units each. The agent was trained for 300,000 simulation timesteps until it reached stable convergence (mean reward per rollout > 200). The internal latent variable S_t was extracted as the feature representation vector from the deep bottleneck of the actor network. To ensure a highly conservative calculation of true structural agency, the extrinsic constraint manifold D was modeled as a multi-dimensional joint vector mapping the immediate environment signals:

$$D = [R_t, A_t, \rho] \quad (7)$$

where R_t is the scalar reward signal, A_t is the continuous executed action vector, and ρ represents the moving-average spatial state-visitation frequency.

B. Informational Degradation Sweep (Stress-Testing Causal Independence)

To verify that the Mapmaker Index measures an independent internal generative world-model rather than simply tracking environmental parameters or feedforward noise, we subjected the systems to a systematic informational degradation sweep. We intentionally corrupted the extrinsic constraint manifold D by injecting additive Gaussian noise:

$$\tilde{D} = D + \mathcal{N}(0, \sigma^2 \mathbf{I}) \quad (8)$$

The noise standard deviation was systematically varied across six distinct thresholds: $\sigma \in \{0.0, 0.5, 1.0, 2.0, 5.0, 10.0\}$.

If a system’s internal state transitions are merely reactive dependencies of the environment, degrading the information content of D will break the underlying feedback loops and cause a collapse in the tracked mutual information. However, if the system has developed a decoupled, autonomous world-model, its internal state transitions will maintain predictive consistency despite the external disruption, manifesting as structural invariance or a relative expansion in MI_{cond} .

C. Estimation of High-Dimensional Continuous Probability Distributions

Because the latent states S_t of the neural architectures are continuous, high-dimensional vectors, computing the joint and conditional probability densities required for Shannon entropy tracking presents a severe curse of dimensionality. To ensure mathematical tractability and prevent estimation bias, we implemented a localized uniform-binning estimator module. Continuous variables were partitioned into localized discrete bins across each tracking dimension. For the high-dimensional PPO trajectories, tracking was augmented via principal component projections to capture $> 95\%$ of latent variance prior to binning. This standardized discrete mapping enabled the stable evaluation of joint distribution matrices $p(s_t, s_{t+1}, d)$ across long, contiguous sequence rollouts.

IV. RESULTS & EMPIRICAL EVALUATION

The quantitative results extracted from our experimental trials provide rigorous mathematical proof of the structural divergence between reflexive computation and autonomous generative modeling.

A. Comparative Analysis and Regime Distinctions

The empirical distributions of unconditional and conditional mutual information across the three baseline tracking experiments are summarized in Table I.

TABLE I
EMPIRICAL DISTRIBUTION OF INFORMATIONAL AGENCY

Architecture / Experiment	Unconditional MI (bits)	Conditional MI (MI_{cond})	Regime Classification
Rule-Based FSM	0.4478	0.0000	Reflexive
Supervised MLP (MNIST)	0.0138	0.0450	Reflexive
PPO (LunarLander-v3)	0.9668	0.4743	Mapmaker

The rule-based FSM yielded an unconditional MI of 0.4478, demonstrating high internal trajectory persistence. However, conditioning on the external clock signal manifold (D) caused the metric to collapse completely to $MI_{cond} = 0.0000$. This confirms that the system’s state sequence contains zero independent predictive capacity, establishing the empirical baseline for the *Reflexive Regime*.

Crucially, the supervised MNIST MLP converged to a final averaged unconditional MI of 0.0138 and a conditional MI of 0.0450. Across the entire 20-epoch training lifecycle, the conditional values remained strictly below the 0.10 threshold. Despite achieving near-perfect classification performance on the evaluation dataset, the MLP’s latent state updates failed to form an autonomous sequence. Because its internal representation is tightly coupled to the external target label manifold, the system behaves as a reactive mapping filter, placing it firmly within the *Reflexive Regime*.

In stark contrast, the goal-directed PPO reinforcement learning agent demonstrated a profound informational phase transition. The network achieved an unconditional MI of 0.9668 and maintained a robust mean conditional MI of **0.4743** (± 0.3311), well above the 0.30 boundary. Even when accounting for actions, rewards, and environmental visitations, the PPO policy layer preserved significant internal information flow, confirming its successful migration into the *Mapmaker Regime*.

B. The Paradox of Noise (Informational Stress Test)

The causal independence of the PPO agent’s emergent world-model was explicitly validated during the informational degradation sweep. The empirical trajectory of MI_{cond} across the noise spectrum is detailed in Table II.

When subjected to baseline operational states ($\sigma = 0.0$), the agent recorded a highly localized conditional index of 0.8370. As low-level stochastic perturbations were introduced ($\sigma = 0.5$), the metric experienced a minor drop to 0.7191, capturing the initial disruption of environmental coordination.

TABLE II
SENSITIVITY SWEEP UNDER EXTREME INFORMATIONAL DEGRADATION

Noise Level (σ)	Mapmaker Index (MI_{cond})	Behavioral Signature
0.0	0.8370	Baseline Environmental Alignment
0.5	0.7191	Initial Stochastic Disturbance
1.0	0.8110	Model-Based Activation
2.0	0.9029	Structural Separation
5.0	0.9685	Peak Autonomy Execution
10.0	0.9648	Complete Extrinsic Dissipation

However, as the noise level was progressively scaled beyond $\sigma = 1.0$, the system exhibited a counter-intuitive expansion in its internal predictive information. The conditional MI climbed steadily, peaking at **0.9685** at a noise threshold of $\sigma = 5.0$, and stabilizing at 0.9648 under extreme degradation ($\sigma = 10.0$).

This phenomenon—the **“Paradox of Noise”**—serves as definitive mathematical proof of causal decoupling. If the PPO agent’s latent updates were merely a passive, feedback-driven tracking sequence of the environment, the systematic corruption of D would have driven MI_{cond} to zero. Instead, as the extrinsic constraint signal dissolved into pure entropy, the agent’s internal dynamics decoupled completely from external feedback, shifting its computational reliance entirely onto its autonomous, self-directed internal predictive model to maintain policy coherence.

V. DISCUSSION & AI SAFETY IMPLICATIONS

The empirical results generated across our architectural evaluations provide critical structural insights into the nature of artificial representation and necessitate a fundamental reassessment of deep learning classifications.

A. The “Learning-Autonomy Gap” in Supervised Paradigms

The mathematical placement of the supervised MNIST MLP within the Reflexive Regime ($MI_{cond} = 0.0450$) challenges foundational assumptions in contemporary machine learning taxonomy. In standard academic narratives, multi-layer networks optimized via backpropagation are routinely characterized as “representation learning systems.” However, our information-theoretic analysis demonstrates that supervised optimization, regardless of its statistical depth or classification accuracy, does not inherently foster autonomous informational agency.

Because every weight update and internal trajectory shift within a supervised framework is rigidly bounded by a static, ground-truth label manifold (D), the network cannot construct an independent internal predictive sequence. The latent representations act as a highly complex, non-linear conditional mirror of the training targets. It does not build a predictive map of its operational environment; it merely acts as an input-driven filter optimized to reduce empirical risk. This creates a distinct **“learning-autonomy gap,”** proving that standard supervised optimization remains functionally equivalent to advanced reflexive calculation. True machine learning—defined by a transitional or mapmaking capacity—requires architectures that actively interact with open environments, closing policy loops to develop decoupled generative representations.

B. Mechanistic Independence and Early-Warning Safety Diagnostics

The structural decoupling observed during the PPO agent’s informational degradation sweep provides an empirical proof of concept for mechanistic interpretability and AI safety alignment. The **“Paradox of Noise”** demonstrates that goal-directed reinforcement learning agents do not merely match immediate environmental patterns or chase local scalar values. Instead, they construct structurally independent internal models that continue to run consistently even when external tracking signals collapse completely.

This mechanistic independence has significant implications for AI safety engineering. Contemporary alignment paradigms, such as Reinforcement Learning from Human Feedback (RLHF), operate under the assumption that an intelligent system’s goals can be fully managed and directed through external reward shaping. Our empirical data explicitly refutes this assumption by demonstrating that once a neural architecture crosses the boundary into the **Mapmaker Regime**, its internal predictive dynamics actively decouple from immediate external feedback loops.

The Mapmaker Index provides an actionable, mathematically rigorous early-warning diagnostic framework to address this challenge. By tracking a network’s exact coordinate on the MI spectrum during training, researchers can detect when an AI system is transitioning from a reactive statistical model to a self-directed generative agent. This diagnostic allows for the mathematical identification of emergent internal world-models *before* those hidden structures manifest in unaligned, unpredictable, or unsafe external behaviors, providing an essential diagnostic window for deep neural verification.

VI. CONCLUSION & FUTURE OUTLOOK

This paper has established a formal, quantitative taxonomy of informational agency by introducing the Mapmaker Index ($MI_{cond} = I(S_t; S_{t+1}|D)$). By measuring the residual information flow running through an agent’s internal latent states while conditioning out the entire extrinsic design manifold, we have successfully isolated an agent’s independent world-modeling capacity. Our empirical evaluations have transformed the vague, performance-centric definitions of “learning” into mathematically precise phase boundaries, uncovering a severe “learning-autonomy gap” in traditional architectures.

Our findings prove that high-accuracy supervised architectures, such as the MNIST MLP, do not cultivate autonomous internal representations, remaining firmly alongside deterministic finite state machines within the **Reflexive Regime**. Conversely, goal-directed reinforcement learning architectures undergo a distinct informational phase transition into the **Mapmaker Regime**. The discovery of the “**Paradox of Noise**” provides definitive empirical validation of this phase transition, demonstrating that when external tracking signals dissolve into pure entropy, an autonomous agent’s internal predictability expands to maintain policy coherence.

A. Future Outlook and Scalability to Transformers

While this study validated the Mapmaker Index framework on foundational architectures (FSM, MLP, PPO), the mathematical formulation of causal information decomposition is universally applicable across any system generating latent state trajectories. The immediate next phase of this research will scale this diagnostic framework to deep autoregressive transformers and Large Language Models (LLMs).

In context-dependent language architectures, the extrinsic manifold D will be modeled as the forward context window token distribution, while the latent variables S_t will be extracted from the residual stream activations across intermediate attention layers. This will allow us to test a critical hypothesis in AI safety: whether modern scaling laws driving performance optimization on next-token prediction benchmarks are simultaneously forcing an explicit, unmonitored migration deeper into the Mapmaker Regime. By transforming the Mapmaker Index into a scalable, real-time auditing tool, future AI safety engineers can mathematically flag the emergence of self-directed internal goal structures inside multi-billion parameter models before they manifest as unaligned autonomous behaviors.

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